ARCHITECTURE JUSTIFICATION

Shashank Arya | PES1UG24CS432 | Problem Statement #04 – Student Project Portfolio & Showcase App

The Student Project Portfolio & Showcase App combines team-owned project evidence, judge-only rubric assessment, and public voting. A layered component architecture gives these workflows a clear presentation boundary while keeping profile/submission, evaluation, voting, ranking, and persistence responsibilities separate. Project records, vote records, and multimedia objects are stored behind distinct interfaces to reflect their different integrity and handling needs.

Voting is the highest-integrity boundary in the supplied requirements. Public Voting requires a verified voter reference and asks the Vote Ledger to atomically record the voter/project pair; a storage-level uniqueness rule supports FR-003 under repeated or concurrent requests. Identity Verification validates the voter rather than trusting portal-supplied credentials. This supports NFR-002 without claiming that identity checks alone eliminate every sybil attempt.

For NFR-001, the leaderboard consumes accepted-vote events only after the ledger commit, refreshes totals from authoritative vote data, and publishes ranking updates to the portal. This separates vote acceptance from ranking refresh without promising an unspecified latency or throughput target. Interfaces remain technology-neutral because no provider, protocol, or deployment topology was supplied.

The component boundaries directly cover FR-001 through FR-005 and the four stated project areas. They also make the identity and duplicate-vote controls visible for comparison with the Jira Bug Tracker. That comparison is pending: the official Lab 3 handout and Scrum, Kanban, and Jira source documents were not in the workspace, so no defect IDs or undocumented constraints are asserted. This is a logical architecture, not evidence of an implemented or tested system.

Source limitation: official handout and Scrum/Kanban/Jira documents were unavailable; reconcile before final upload.